EN5500 - Computer Systems Assignment 01
Real-Time Sensor Data Processing

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May 1, 2026

A real-time data processing system is designed to collect and analyze data from a sensor. The sensor produces a continuous stream of **integer-valued samples in the range 0 to 100**. The data is processed sequentially in **blocks of 100 samples**.
You are required to generate your own input data to simulate the sensor output (either randomly or using a suitable test pattern). For each block of 100 samples, the system computes the following statistical parameters:

- Maximum value
- Minimum value
- Average (mean)
- Standard deviation
- Number of missing samples
- Number of corrupted or invalid samples

(a) Program Design and Implementation [40 Marks]

Write a program in a programming language of your choice (C, C++, Java, Python, etc.) that:

1. Generates sensor data samples in the range 0 to 100
2. Accepts the generated data sequentially as a data stream
3. Groups the incoming samples into blocks of 100
4. Computes the maximum, minimum, average, and standard deviation for each block
5. Displays the computed results after processing each block

(b) Performance Analysis Under High Data Rate [20 Marks]

In a practical real-time system, the rate at which sensor data arrives may increase significantly.

1. Explain the impact of a very high data arrival rate on:
   - Processor utilization
   - Memory requirements
   - System latency
2. Discuss the possible consequences if the system is unable to process data at the incoming rate.

(c) Handling Missing, Corrupted, and Delayed Data [20 Marks]

In real-world sensor systems, data may be unreliable.

1. Explain how the system should handle the following situations:
   - Missing data samples
   - Corrupted or invalid data values
   - Delayed arrival of sensor data
2. Discuss the effect of these issues on the accuracy of the computed statistical parameters.

(d) System Design Improvements [15 Marks]

Suggest **any two techniques** that can be used to ensure reliable and efficient operation of the system under conditions of high data rate and unreliable data.

For each technique, clearly explain how it improves system performance, reliability, or robustness.

(e) Extension [5 Marks]

Implement or propose **any one** of the following enhancements:

- A configurable block size that can be changed at runtime
- A sliding window approach for continuous statistical analysis
- A timeout mechanism to process incomplete data blocks when delayed data is detected

Submission

- Upload a report with name format is "Your index no_EN5500_A01.pdf". Include the index number and the name within the report as well. Include a link in the report that directs to your code, and it should be executable.
- The interpretation of results and the discussion are important in the report.
- Pay careful attention to formatting such as font size, spacing, and margins.
- Include a title page with necessary information (e.g., title, author, date, index no).
- Use consistent and professional formatting throughout the document.
- Plagiarism will be checked and in cases of plagiarism, an extra penalty of 50% will be applied. In case of copying from each other, both parties involved will receive a grade of zero for the assignment. Academic integrity is of utmost importance, and any form of plagiarism[1] or cheating will not be tolerated.
- An extra penalty of 20% is applied for late submission.

[1] https://en.wikipedia.org/wiki/Plagiarism